MIL-C-6O118(MU)
4 JUNE 1964

MILITARY SPECIFICATION

COMPUTER, BALLISTICS: M16

1. SCOPE

1.1 This specification covers one type analog computing instrumentation known as computer Ballistics: M16 (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 —Parts Equipment and Tools for Ordnance Materiel: Packaging of.

MIL-I-45607 —Inspection Equipment, Supply and Maintenance of.

MIL-B-60119 —Ballistics Computer Subassembly: 8635273 (Control Unit).

MIL-B-60120 —Ballistics Computer Subassembly: 8635274 (Zeroing Control Unit).

MIL-B-60121 Ball is tics Computer Subassembly: 8635275 (output Unit).

MIL-B-60122—Ballistics Computer Subassembly: 8635276 (Input Unit).

MIL-B-60123—Ballistics computer Subassembly: 8635370 (computer Unit).

MIL-B-60124—Ballistics Computer Subassembly: 8635412 (Cant Unit).

STANDARDS

MILITARY

MIL-STD-105-Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109-QUalitY Assurance Terms and Definitions.

MIL-STD-202-Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

U.S. Army Munitions Command

F10516500 —Computer Ballistics: M16.

IEL 10516500-List of Inspection Documents for Computer Ballistics: M16.

{Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1220

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3. REQUIREMENTS

3.1 Material s Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication. The computer shall be manufactured in accordance with Drawing F10516500 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements . of Specification MIL--13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 Subassemblies. Each Computer Ballistics M16 submitted in conformance with this specification shall be composed of subassemblies which have complied with all their respective requirements as specified in the following specifications.

Ballistics Computer Subassembly: 8635273 (Control Unit) MIL-B-60119
Ballistics Computer Subassembly: 8635274 (Zeroing Control Unit) MIL-B-60120
Ballistics Computer Subassembly: 8635275 (Output Unit) MIL-B-60121
Ballistics Computer Subassembly: 8635276 (Input Unit) MIL-B-60122
Ballistics Computer Subassembly: 8635370 (computer unit) MIL-B-60123
Ballistics Computer Subassembly: 8635412 (Cant Unit) MIL-B-60124

3.5 Performance.

3.5.1 Computer output cant out With the computer connected in accordance with Drawing F10516500 utilizing an indicating device electrically and mechanically appropriate to indicate the output functions of the computer for reticle displacement, and with a 7 inch pound load applied to the output unit shaft for the ballistics drive and with power applied, the elevation and deflection corrections shall conform to the values and tolerances of Table I for conditions and ammunitions types. The requirements of Table I shall be met with the power supply voltage set at each of the following three voltages, 18 volts, 24 volts and 30 volts dc.

3.5.2 Computer output-cant off.
tion and deflection corrections shall conform to the “zero” degree cant values and tolerances in table I when the cant switch is positioned to the OFF position.

3.5.3 Boresight. Elevation and deflection corrections shall zero within 0.05 mils when the computer is placed in the boresight mode of operation at 1200 meters range input.

3.5.4 Zeroing. Elevation and deflections corrections shall be modified by use of the zeroing knobs in a direct relationship with the knob graduations within a tolerations of .05 mils at the zero graduation mark to within .18 mils at the 3 mil graduation marks.

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TABLE I. *Computer output*

AMMUNITION—APDS—M392

TOTAL CORRECTION IN MILS

INPUT SHAFT Position (CCW) Turns	Degrees	(Informational) Range Meters	0° CANT Elevation	0° CANT Deflection	+15° CANT Elevation	+15° CANT Deflection	−15° CANT Elevation	−15° CANT Deflection
2	334.9	600	2.506	L.424	2.444	R.035	2.473	L.883
8	48.1	1000	3.021	L.064	2.916	R.606	2.963	L.733
9	356.6	1200	3.392	R.052	3.264	R.870	3.324	L.766
12	94.8	1500	4.055	R.143	3.892	R1.183	3.971	L.896
15	69.7	2000	5.274	R.270	5.048	R1.679	5.159	L1.137
19	116.5	3000	8.008	R.473	7.652	R2.641	7.830	L1.694
22	91.5	4000	11.140	R.677	10.643	R3.555	10.884	L2.198
		AMMUNITION—HEP—M393						
2	334.9	600	7.440	L.206	7.142	R1.753	7.304	L2.165
8	48.1	1000	12.209	R.338	11.681	R3.492	11.946	L2.813
9	356.6	1200	15.107	R.571	14.447	R4.550	14.778	L3.406
12	94.8	1500	19.942	R.894	19.050	R6.125	19.498	L4.331
15	69.7	2000	29.866	R1.422	28.522	R9.205	29.193	L6.354
17	311.8	2600	44.970	R2.256	42.947	R13.937	43.955	L9.413
18	352.2	2900	54.080	R2.678	51.614	R16.698	52.843	L11.332
		AMMUNITION—HEAT—M456						
2	334.9	600	3.496	L.446	3.414	R.360	3.414	L1.251
8	48.1	1000	4.966	L.104	4.810	R1.149	4.810	L1.357
9	356.6	1200	5.930	0.000	5.745	R1.551	5.745	L1.551
12	94.8	1500	7.595	R.068	7.350	R2.080	7.350	L1.944
15	69.7	2000	11.030	R.155	10.664	R3.112	10.664	L2.801
19	116.5	3000	21.398	R.242	20.674	R5.868	20.674	L5.382
22	91.5	4000	39.912	R.287	38.559	R10.933	38.559	L10.357

TOLERANCES ON VALUES OF TABLE I

ELEVATION

450 METERS TO 2000 METERS ±.15 MIL
2000 METERS TO 4700 METERS ±.20 MIL

DEFLECTION

450 METERS TO 4700 METERS ±.20 MIL

3.5.5 *Effective full charge corrections (EFC).* Elevation corrections shall be increased by the percentages listed in table II as the EFC dial is advanced to the values listed. An additional tolerance of 0.1 mil is allowed on elevation corrections above zero EFC.

3.5.6 *Effective full charge count.* Actuation of the EFC switch unit shall cause the EFC dial to advance 4 full counts on APDS–M392–A1, one full count on APDS–M392–A2; 1/10 count on HEP–M393 and 2/5 count on Heat–M456 for each complete cycle of the switch. The roller arm of EFC switch to be held down a minimum of 2 seconds for each test cycle.

TABLE II.

Effective full charge corrections

EFC Dial Setting	Percent Correction Ammunition Type APDS–M392	HEP–M393	Heat—M456
0	0.000%	0.000%	0.000%
100	1.421%	3.080%	0.213%
200	2.842%	5.617%	0.426%
300	4.264%	7.826%	0.638%
400	5.685%	9.808%	0.848%

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3.5.7 Servo stability. The tachometer adjustments shall control the dynamic response of both the elevation and deflection servos from under damped (oscillation and/or excessive overshooting in the full counterclockwise position) to over damped (sluggish with no overshoots in the full clockwise position).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and test shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and subassemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. definitions of inspection terms not otherwise defined herein shall be as 1isted in Standard MIL-STD-109.

4..2 Inspection provisions.

4.2.1 Inspection lot. Unless otherwise specified by the contracting officer, inspection lot size, formation and presentation of lots shall be in accordance with ‘Submission of product” as specified in Standard MIL-STD-105.

4.2.2 Examination and tests. Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL-F-13926 and the Supplementary Quality Assurance Provisions listed in the List of Inspection Documents, IEL-10516500. Examinations and tests related to section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105. and the inspection level and sampling plan specified in table III titled, “Classification of Defects”. Examinations and tests for packaging, packing, and marking shall be in accordance with Specification MIL-P-14232 and section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance of rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

TABLE III. *Classification of defects*

Use Level II of Table I with Sampling Plan Table II-A of Standard MIL–STD–105

	Requirement	Test procedure
Critical: None.		
MAJOR: AQL 1.5		
101. Computer output (cant on)	3.5.1	4.5.2
102. Computer output (cant off)	3.5.2	4.5.3
103. Boresight	3.5.3	4.5.4
104. Zeroing	3.5.4	4.5.5
105. EFC Corrections	3.5.5	4.5.6
106. EFC Count	3.5.6	4.5.7
107. Servo stability	3.5.7	4.5.8
MINOR: None.		

Note. The above tests shall be performed at standard ambient temperature (60° to 90°F.).

4.2.3 Disposition. of nonconforming product. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection” as specified in Standard MIL-STD-105.

4.3 Control tests. One Computer M16 shall be selected at random from each one hundred produced. or from each month’s

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production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test.	Requirement	Test Procedure
301. Materials	3.1	as per specification; drawing
302. Fabrication	3.2"~	Visual–per drawings
303. General specification	3.3	Visual–MIL-F-13926

4.3.1 Control test failure. Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in Specification MIL-I-145607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL-F-13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90 % of the prescribed tolerance limit. construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 Inspection equipment accuracy requirements. The devices required in the test methods and procedures herein, to measure the values specified in Section 3, shall provide readings to the accuracy specified below

Item number	Description
1. Input Turn indicator	Indicate turns of input shaft to 0.1 degree and to the positions specified in table I.
2. E.F.C. Switch Actuation Counter	Indicate E.F.C. O to 400 switch actuation with 2 second minimum time delay.
3. Cant Unit Positioner	Position cant unit with mounting surface horizontal (0') and at $\pm 15^{0}$ of cant within O. 1 degree.
4. Elevation Indicator	7 inch pound load and a protractor dividing head attached to Ballistics Drive output Shaft, Revolution Counter Dial graduated at 5 mils per revolution to a precision of 0.01 roil.
5. Deflection Test Indicator	Electrically and mechanically specified in figures 1 and 2.
6. A.C. Power Supply	115 volts rms $\pm$ 5% , 400 Cps $\pm$ 1%, 4% maximum harmonic distortion, 1.5 % maximum change in harmonic distortion.
7. D.C. Power Supply	18 to 30 volts maximum ripple, 0.5 volt peak to peak, 20 ampere output.

4.5 Test methods and procedures

4.3.1 Performance tests. The Ballistics Computer: M16 shall be subjected to the following tests using the items of test equipment specified in 4.4.2. Prior to tests, the inspector shall ascertain that each subassembly of the computer has been inspected and found to be in conformance with their respective specifications listed in 3.4.

4.5.2 Computer output (cant on). To determine conformance to 3.5.1 the computer shall be connected to the power supplies (items 6 and 7) and 18 V13C supplied to the system. The deflection test indicator (item 5) shall be attached to the computer unit. Attach the input turn indicator (item 1) to the input shaft of the input unit such

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that the indicator reads 356.6 degrees from the zero position counterclockwise, when the 1200 meter index on the input unit range dial and the scribe lines of the input shrift and input unit housing are in coincidence. Actuate APDS on a control unit (Dwg. F8635273) and set the cant unit mounting surface to horizontal (0°). Allow the computer system to come to a solution for 600 meter input Observe the elevation and deflection outputs for conformance to Table I. Repeat the procedure with the HEAT button of the control unit, and finish with the HEP button, (Warning: If the button is not held down for a sufcient length of time incorrect ammo selection can occur.) This completes the 600 meter, 0° cant test. Continue testing at O° cant for the remaining ranges of table I. Repeat the tests for all ranges at +15 ° of cant and again for all ranges at -15° cant. This concludes an 18 VDC check. Repeat the entire procedure at 24 VDC and 30 VDC.

4.5.3 Computer output (cant off). To determine conformance to 3.5.2 proceed as in 4.5.2 for all ammos and all ranges but at the 0° cant setting only. This test may be conducted in conjunction with the first part of 4.5.2 by switching the cant unit off after each 0° solution observed, and returning it to the "on" position before proceeding to the next ammo or range.

4.5.4 Boresight. To determine conformance to 3.5.3 open. the zeroing unit and pull out the boresight knob. Set the range to 1200 meters, and observe elevation and deflection solutions for zero within the tolerance specified.

4.5.5 Zeroing. To determine conformance to 3.5.4, after determining that each knob can zero within .05 mils. select one knob and note its correction variation from zero. Rotate the knob to the 1 mil position. The correction shall have moved one mil with a tolerance of .09 mils. Rotate the knob to 2 mils and observe the correction as having moveed 2 mils from zero with a tolerance of .14 mil. Lastly, move the knob to the three mil position and note the correction as having moved 3 mils, from its zero position, within .18 roils. repeat the procedure with each zeroing knob.

4.5.6 Effcctivc full charge corrections (EFC). To determine conformance to 3.5.5 open the computer unit cover and advance the EFC dial to) the 100 mark and apply the percentage correction factor of table II to the table I values. With this new table, proceed to test as in 4.5.3. Make new tables for 200, 300, and 400 EFC settings and test from these tables as in 4.5.3. The count test of 4.5.7 may be conducted in conjunction with this test.

4.5.7 Effective full-charge count. To determine conformance to 3.5.6 actuate the EFC switch 25 actuations with APDS-A1 ammo indicated on a control unit and EFC assembly. The dial shall advance to 100. Actuate the EFC switch 100 times with a control unit switched to APDS-A2. The E.F.C. dial should advance 25 units. Proceed for 100 and 25 actuations for the HEP and the HEAT ammos respectively. The dial should advance 10 counts for each of these ammos. During the test, care shall be taken to hold the EFC roller down for at least 2 seconds during each actuation.

4.5.8 Servo stability. To determine conformance to 3.5.7, make this test when the computer is slewing (in the process of going to an answer). Open the computer unit cover and rotate the tachometer pots counterclockwise and observe the elevation and outputs for oscillation. Increase rotation of the pots in a clockwise direction until the performance becomes sluggish, to the point of no overshoot. Return the pots to their original position of minimum overshoot.

5. preparation fOR DELIVERY

5.1 packaging. packing anti marking. Packaging, packing and marking shall be in accordance with Specification MIL-P-14232

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to the applicable packaging data sheet. he level of protection shall be as specified the procurement document (see 6.2).

6. NOTES.

6.1 Intended use. The Ballistics Compur: M16 is an analog computing instrument sed to receive the log of range data from a imagefinder calibrated for range measurements in meters, and computer elevation and deflection corrections for various ammunitions and transmits same to a gun sighting system in the 105MM Gun, Tank M60A1.

6.2 Ordring data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging and packing (see 5.1).

(c) Applicable Packaging Data Sheets Number (see 5.1).

(d) Applicable stock number.

Custodian:
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Project No. 1220-a160Y

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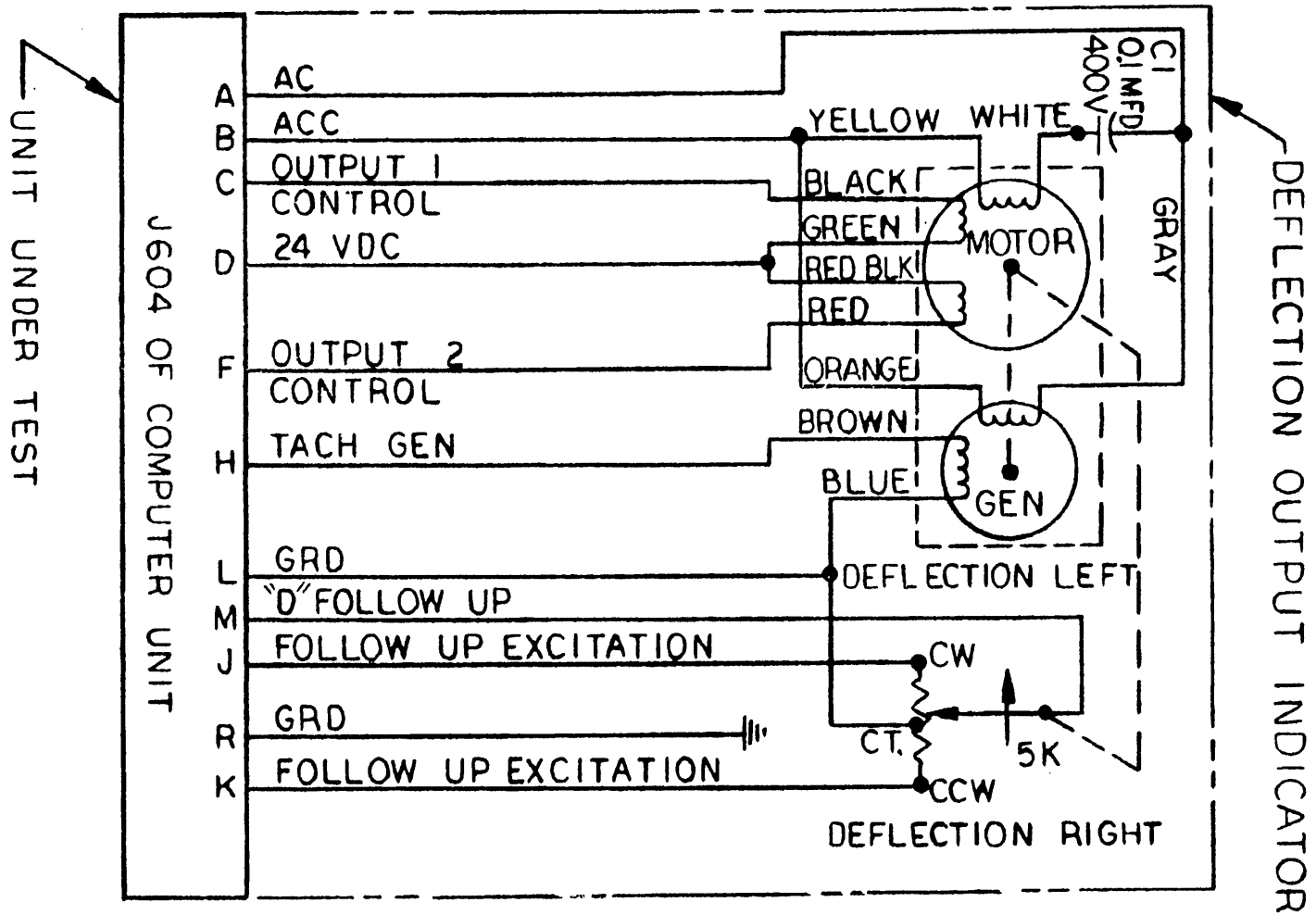
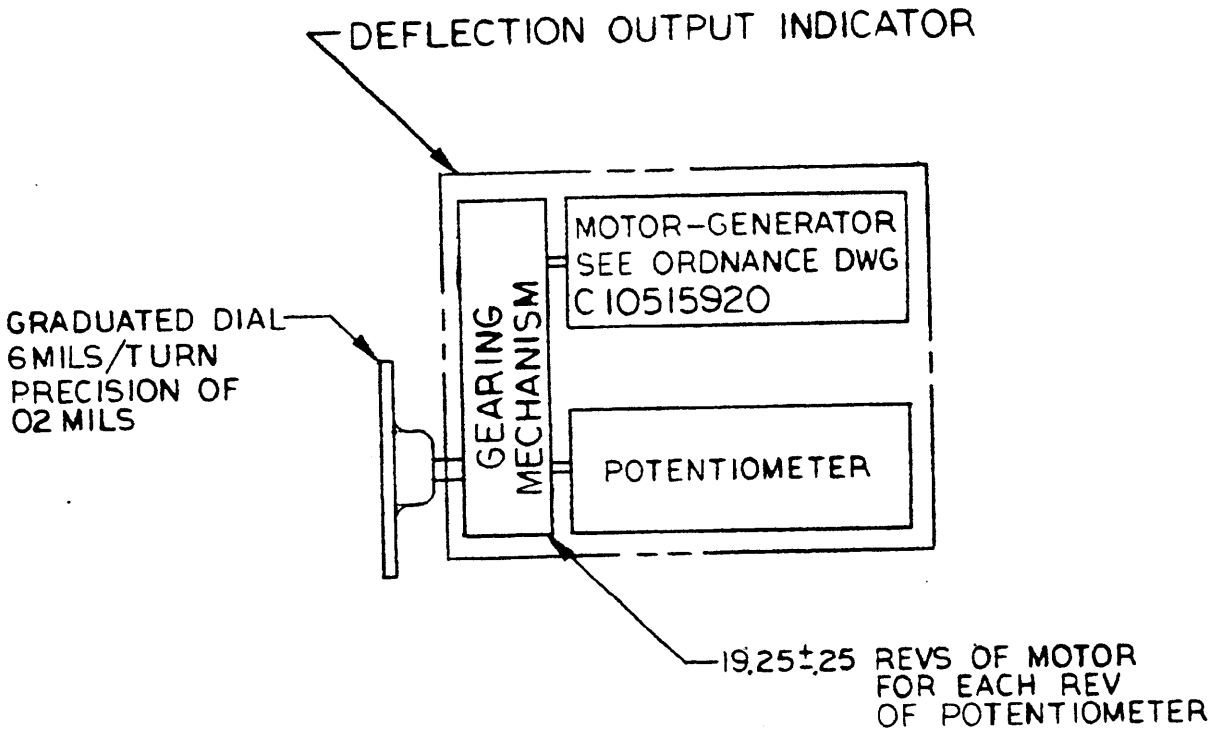


FIGURE 1.

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RESPONSE TIME TO 22.0 MILS/SEC SLEW SPEED NOT TO EXCEED .073 SECONDS. (NOMINAL TOP SLEW SPEED 35.0 MIL/SECOND)

POTENTIOMETER.
ELECTRICAL CHARACTERISTICS

3600 DEG ± 1 DEG FROM CW TO CCW TERMINALS
CENTER TAP AT ELECTRICAL CENTER WITHIN ± 1 DEG.
LINEARITY ±.025 %
5000 OHMS ± 5 %
4500 STEPS OF RESOLUTION MINIMUM

FIGURE 2.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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